

AMENDMENT NO. 1 JULY 1994
TO
IS 5613 (Part 2/Sec 1) : 1989 CODE OF PRACTICE FOR
DESIGN, INSTALLATION AND MAINTENANCE OF
OVERHEAD POWER LINES
PART 2 LINES ABOVE 11 kV AND UP TO AND INCLUDING 220 kV
Section 1 Design
(First Revision)

(Page 4, clause 3) — Insert the following and renumber subsequent clause.

'4.2 For the safety requirement of low flying Military Aircrafts, all the transmission lines and transmission line structures falling within the safety zone of airfields and air to ground firing ranges shall meet the requirements of Directorate of Flight Safety, Air Headquarters given in Annex A '

(Page 29, Annex A) — Insert the following Annex are and remember the subsequent Annexes

ANNEX A
(Clause 4 2)

VISUAL AIDS FOR DENOTING TRANSMISSION
LINES AND TRANSMISSION LINE STRUCTURES —
REQUIREMENT OF DIRECTORATE OF FLIGHT
SAFETY

All the Power Utilities shall comply with the following :

- a) The transmission lines and transmission line structures of height 45 maad above shall be notified to the Directorate of flight Safety (DES), Air Headquarters (Air HO), New Delhi.
- b) For construction of any transmission line/structure or a portion there of falling within a radius of 20 km around the Defence aerodromes and air to firing ranges provisions of the Aircraft Act 1934, Section 9A as amplified by the associated Gazette Notification SO 988 Part 11, Section 3, Subsection (11) dated 1988-03-26 shall be complied with lowards

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this, a No Objection Certificate - (NOC) shall be obtained from the concerned aerodrome authorities.

- c) Within a radius of 10 km around aerodromes and air to ground firing ranges, all transmission lines and structures of height 45 meters or more shall be provided with day and night visual aids.
- d) In all other areas, outside a radius of 10 km from aerodromes, only those portions of transmission lines and structures of any height identified to pose a hazard to aircraft by the Directorate of Flight Safety shall be provided with day visual aids.

2. DESCRIPTION OF VISUAL AIDS

2.1 Day Marking

- i) *Line Markers* — Coloured globules of 40-50 cm diameter made of reinforced fibre glass or any other suitable material, weighing not more than 4.5 kg each with suitable clamping arrangement and drainage holes shall be installed on the earth wire(s) in such a manner that the top of the marker is not below the level of the earth wire. Up to 400-metre span, one globule shall be provided in the middle of the span on the highest earth wire. In case of double earth wires, the globule may be provided on any one of them. For span greater than 400-metres, one additional globule may be provided for every additional 200-metre span or part thereof. Half orange and half white coloured globule should be used. A typical sketch of marker is given in Fig. 1.
- ii) *Structure Marking*—The structure portion excluding cross-arms above 45 in height shall be painted in alternate bands of international orange and white colours. The bands shall be perpendicular to the vertical axis and the top and bottom bands shall be orange. There shall be an odd number of bands. The maximum height of each band shall be 5 m. A typical sketch of marking a structure is given in Fig. 2.

2.2 Night Marking

2.2.1 Medium and low intensity obstacle lights as per Fig. 3 on a complex obstacle such as towers supporting overhead wires should have a night time intensity as per ICAO requirements in International Standards recommended practices. The light on top of the structure should flash at the rate of 20 sequences per minute.

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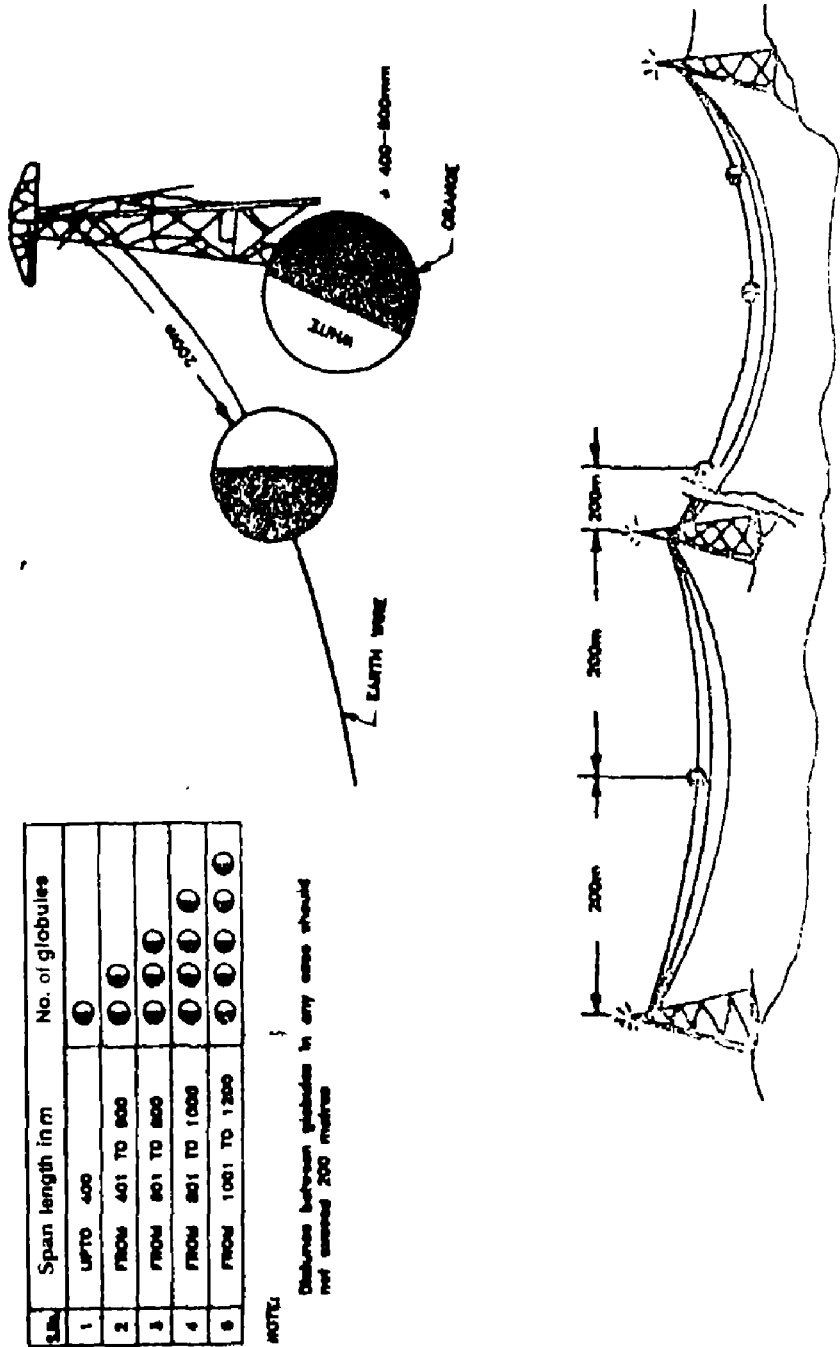


Fig. 1. MARKING WITH GLOBES

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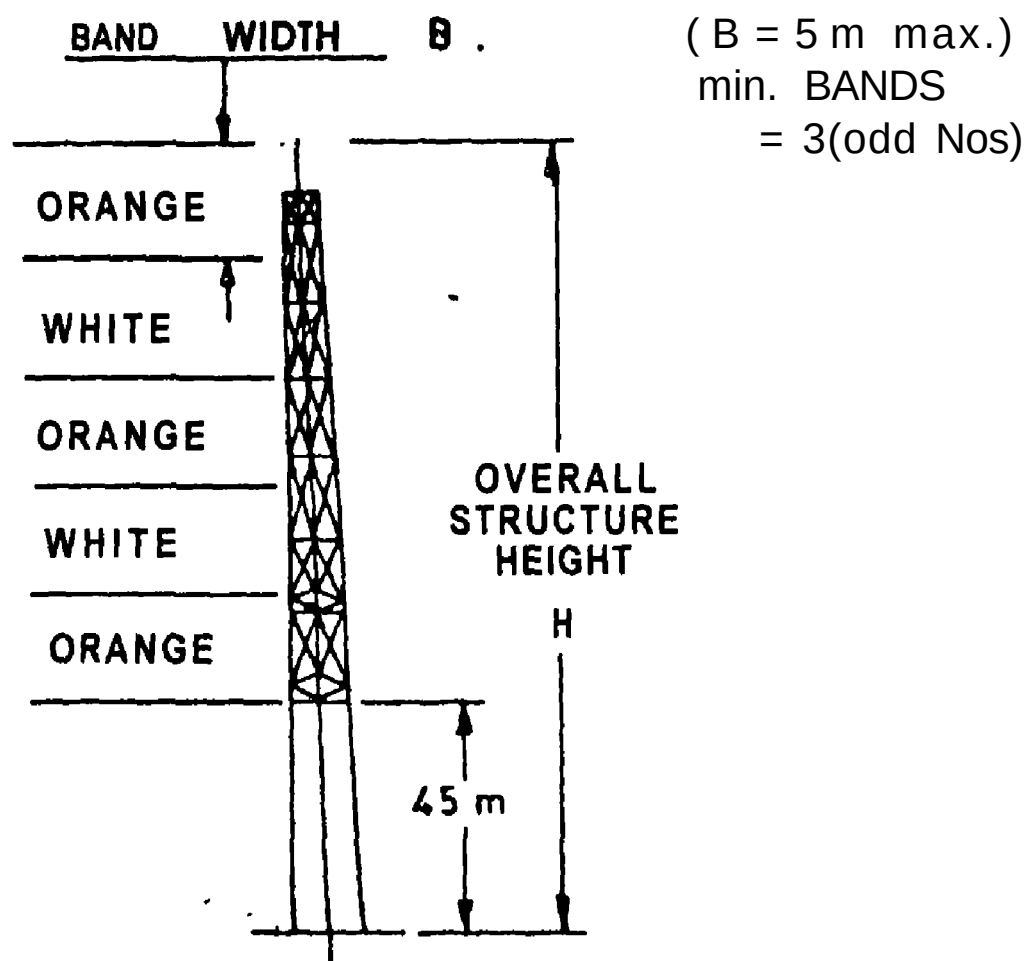
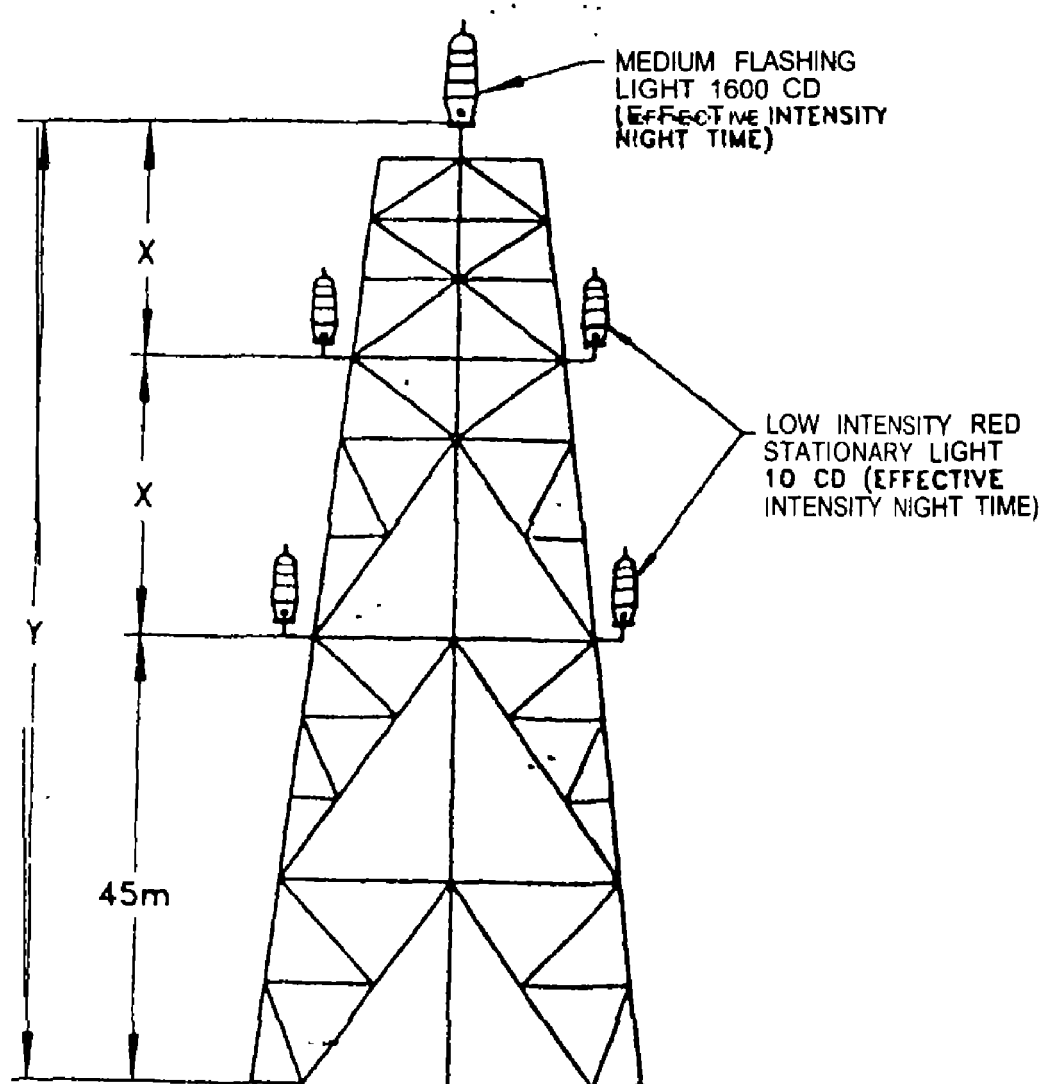


FIG. 2 DAY MARKING ALTERNATE BANDS

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$$\text{Number of Lights} = N = \frac{Y \text{ (METRE)}}{45}$$

$$\text{Light Spacing} = X = \leq 45$$

FIG. 3 NIGHT MARKING OF POWER TRANSMISSION TOWER

(ETD37)